

Day 3 – Network Topologies

Aim

To understand and implement different network topologies using Cisco Packet Tracer.

Objective

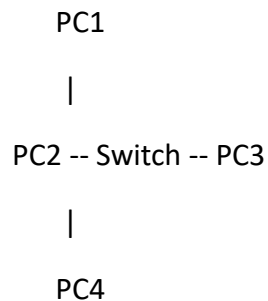
- To study different types of network topologies.
 - To build Star, Bus, Ring, and Mesh topologies in Cisco Packet Tracer.
 - To understand the advantages and disadvantages of each topology.
 - To compare their reliability and fault tolerance.
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Topology 1: Star Topology

Description

In Star Topology, all devices are connected to a central device such as a switch.

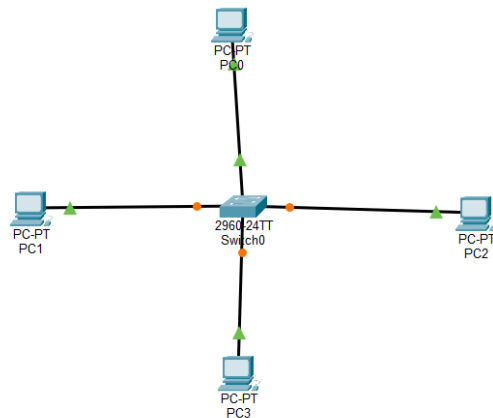
Diagram



Procedure

1. Open Cisco Packet Tracer.
2. Add:
 - 1 Switch
 - 4 PCs
3. Connect all PCs directly to the switch using Copper Straight-Through cables.
4. Verify the connections.

Screenshot 1: Star Topology



Observation

- All devices communicate through the switch.
- Failure of one cable affects only one device.
- Failure of the switch affects the entire network.

Topology 2: Bus Topology

Description

In Bus Topology, all devices share a common backbone cable.

Diagram

PC1 ---- PC2 ---- PC3 ---- PC4

Procedure

1. Create a visual representation of Bus Topology.
2. Arrange PCs in a straight line.
3. Connect devices to represent a shared communication path.

Screenshot 2: Bus Topology



Observation

- All devices share the same communication medium.
 - Backbone cable failure can affect the entire network.
 - Troubleshooting is difficult.
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Topology 3: Ring Topology

Description

In Ring Topology, devices are connected in a circular path.

Diagram

PC1 ---- PC2

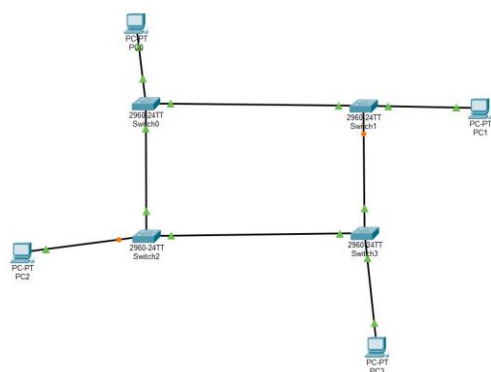


PC4 ---- PC3

Procedure

1. Add four PCs.
2. Connect them in a circular fashion.
3. Ensure the ring is complete.

Screenshot 3: Ring Topology



Observation

- Data travels through the ring.
- A break in the ring can disrupt communication.
- Expansion is more difficult than Star Topology.

Topology 4: Mesh Topology

Description

In Mesh Topology, every device is connected to every other device.

Diagram

A-----B

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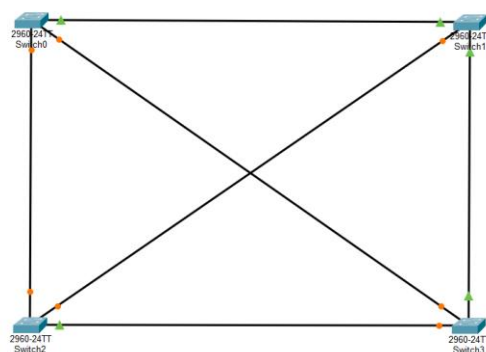
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C-----D

Procedure

1. Add four PCs.
2. Connect each PC to every other PC.
3. Verify all links are established.

Screenshot 4: Mesh Topology



Observation

- Multiple communication paths are available.
- Network remains operational even if one path fails.

- Highly reliable but expensive.

Comparison of Topologies

Topology Central Device Reliability Fault Tolerance Cost

Star	Yes	High	High	Medium
Bus	No	Low	Low	Low
Ring	No	Medium	Medium	Medium
Mesh	No	Very High	Very High	High

Result

All four network topologies (Star, Bus, Ring, and Mesh) were successfully created in Cisco Packet Tracer and their characteristics were observed.

Conclusion

The experiment helped in understanding how different network topologies are structured and how they affect network reliability, scalability, fault tolerance, and troubleshooting. Star Topology was found to be the most commonly used topology in modern LANs, while Mesh Topology provides the highest reliability.